

Varshap Walia

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Technical Skills

- **Languages:** Java, Python, C#/.NET, Go, JavaScript
- **AI & Backend:** LLM Agents, RAG, LangGraph, LlamaIndex, FAISS, PyTorch, Kafka, Redis, gRPC, OAuth, OIDC, SAML
- **Frameworks & Databases:** .NET/ASP.NET, Spring Boot, Django, FastAPI, PostgreSQL, MongoDB, MySQL, Elasticsearch
- **Cloud & Tools:** AWS, GCP, Kubernetes, Docker, Helm, Terraform, GitLab CI/CD, GitHub Actions, Datadog, New Relic, Sentry

Experience

Software Developer

Jan. 2025 – Present

Thales Group

Ottawa, ON

- Developed and scaled distributed **IAM microservices** in **Java** and **.NET** supporting **SAML, OAuth, and OIDC** across multi-tenant environments on **Kubernetes** and **GCP**, serving **1M+ users**.
- Reduced p95 latency in critical **IAM workflows** from **~800ms** to under **120ms (~6× improvement)** through query optimization, indexing, and distributed **Redis** caching.
- Built **autonomous CI/CD self-healing system** in **Python** using **LangGraph LLM agents**, automatically resolving **~35–40%** of flaky test failures without human involvement.
- Built **RAG-based incident analysis platform** integrating Jira, logs, traces, and metrics using **FAISS retrieval**, reducing MTTR by an estimated **20–30%**.
- Led **LLM-based .NET 5 → .NET 8 migration** with Windows → Linux transition, achieving **~82%** automation and saving **~\$1M** in licensing costs.
- Built distributed **load testing framework** using **Kubernetes, Helm, Terraform, and Locust** on **GCP**, simulating **100K+ concurrent users** and surfacing bottlenecks before production deployment.

M.Sc. Applied Computer Science (AI)

Sept. 2022 – Dec. 2024

Concordia University

Montreal, QC

- Teaching Assistant for graduate-level **Applied AI and Distributed System** courses while completing degree full-time.
- Interned twice at **Thales Group** during the program, **converting to full-time** Software Developer role upon graduation.

Software Engineer II

Feb. 2022 – Aug. 2022

Vahak

Bengaluru, IN

- Redesigned geolocation platform built on **Go, Elasticsearch, and OpenStreetMap data**, expanding points-of-interest coverage by **4×** while improving data integrity and matching accuracy.
- Improved search relevance by **~89%** through relevance tuning and query analysis, increasing booking conversion.
- Built **Kafka-based pipelines** processing **~2M events/day** into MongoDB and S3 for analytics and ML workflows.
- Mentored junior engineers on backend design patterns, improving code quality and reducing review cycle time.

Software Engineer

Apr. 2019 – Jan. 2022

Pinkblue

Bengaluru, IN

- Identified manual handoff steps inflating delivery error rates; redesigned logistics workflow with **automated validation gates**, reducing errors **~17%** and saving an estimated **\$187K annually**.
- Architected and built **POS and logistics platform** using **Python, Django, PostgreSQL, and AWS** across **100+ hospitals and 450+ pharmacies**, unifying inventory, order processing, and fulfillment as the **first system to span both verticals**.
- Led **GCP → AWS migration** leveraging **EC2, S3, EKS, MSK, SNS, Redshift, and OpenSearch**, reducing downtime incidents by **~23%** and improving observability with **New Relic and Sentry**.
- Built **real-time pricing engine** using market and vendor data feeds, improving order conversion by **~15%**.

Backend Engineer

Nov. 2017 – Mar. 2019

Inclov Technologies

Gurgaon, IN

- Built **encrypted messaging and push notification systems** on **Firebase** from scratch; improved delivery reliability from **72% → 94%** and increased engagement by **~41%** over six months post-launch.
- Built **ML-based content moderation pipeline** cutting moderation workload **~27%**; redesigned onboarding for **WCAG 2.1 compliance**, improving activation rates **20–25%**.

Education

Concordia University

Montreal, QC

Master of Applied Computer Science, **Major in AI**

Sept. 2022 – Dec. 2024

Kurukshetra University

Kurukshetra, IN

Bachelor of Technology in Computer Science

Aug. 2013 – Aug. 2017